Project Design Document: Scurr-E!

1. Design Intent

- *A fun, interactive game with fast paced adrenaline, immersiveness, and escaping capture in time. It is a physical, real-world robot game where players must chase and tag a fleeing WALL-E robot before a countdown timer expires. The robot actively tries to escape, reacting to the presence of nearby players and fleeing at speed. Players must physically catch the robot and press a button to defuse the bomb before the NeoPixel ring timer counts to zero.*

- *This game is for people of all ages*

2. Inspiration & References

- *Existing products / ideas:*

  - *Mars robotic kiosk that chases customers to hand out samples at retail/grocery stores*
  - *Think & Learn Teach 'n Tag Movi: A popular, interactive robot designed for young children, offering three game modes including tag.*
  - *Robomates: Open-source, mini, two-wheeled, self-balancing robots designed for interactive tabletop games such as tag, football, and push battles*

- *Images / links*

  - *WALL-E Robot: https://in.pinterest.com/pin/1130614681510736415/*
  - *Mars Bar Rover: https://gizmodo.com/this-mars-bar-rover-will-chase-you-around-a-store-and-t-1846514705*
  - *Robomates tag: https://www.reddit.com/r/robomates/comments/1n5put6/robomates_tag_our_first_game_with_mini_robots/*

3. User Journey

Step-by-step experience of the user:

1. *User starts… by opening the MIT App Inventor app on their phone*

2. *The user interacts…by selecting a difficulty level — Easy, Medium, or Hard. Then, the user presses the "Start" button.*

3. *The system responds… when the robot's NeoPixel ring counts down 5 seconds (flashing red), then turns green and the robot begins wandering. The robot wanders slowly and unpredictably while the player is far away. The arms wave lazily. The NeoPixel ring glows a full steady green. As the player gets within ~3 metres, the PIR motion sensor triggers. The robot panics- it spins and bolts in a random direction. The arms start to flail. The ticking sound speeds up. The NeoPixel ring begins shifting from green toward amber toward red as the timer counts down. The player must physically chase the robot and press the push button on it to defuse it. If they succeed- the robot plays a victory fanfare, the NeoPixel ring flashes green, and the arms raise in celebration. If the timer hits zero before the player tags it- the robot plays an explosion sound, the NeoPixel ring flashes red rapidly, and the arms flail wildly. The player loses. The robot also has fun, interactive features like waving hi, doing a mini dance, etc., outside of the game modes*

4. Success Criteria

Define measurable success:

- *Basic functionality:*
  - *'On' button switches the robot on*
  - *'Start' button lets the player choose the game mode*
  - *Upon choosing the game mode, the neopixel turns from red to green and the robot becomes 'alive' in 5 seconds*
  - *The robot shuts down when the push button is pressed*

- *User experience*
  - *Easy to use- The button on the robot can be pushed easily*
  - *Intuitive- The MIT App interface is understood with ease*

- *Tracking efficiency*
  - *Ability to detect humans using PIR sensor and driving away*

- *Obstacle avoidance*
  - *Ability to detect obstacles to avoid bumping into them*

- *Mobility*
  - *Can shift between the 3 speed modes*
  - *Ability to drive at each speed mode with ease*

- *Reaction Time*
  - *Ability to quickly detect a sudden change in direction by the users*

5. System Overview

Difficulty is set from the MIT App Inventor interface before each round and affects three parameters simultaneously:

Difficulty	Timer	Wander Speed	Flee Speed	Arm Speed
Easy	90 seconds	Slow	Medium	Slow wave
Medium	60 seconds	Medium	Fast	Moderate flail
Hard	30 seconds	Fast	Very fast	Frantic flail

Bluetooth Commands

App Action	Command Sent	Robot Response
Select Easy	EASY	Sets 90s timer, slow speeds
Select Medium	MEDIUM	Sets 60s timer, medium speeds
Select Hard	HARD	Sets 30s timer, fast speeds
Press START	START	Begins 5s countdown then enters game
Press STOP	STOP	Immediately halts all motors and resets

The robot operates in two distinct behavioural states based on whether the PIR sensor detects a nearby player:

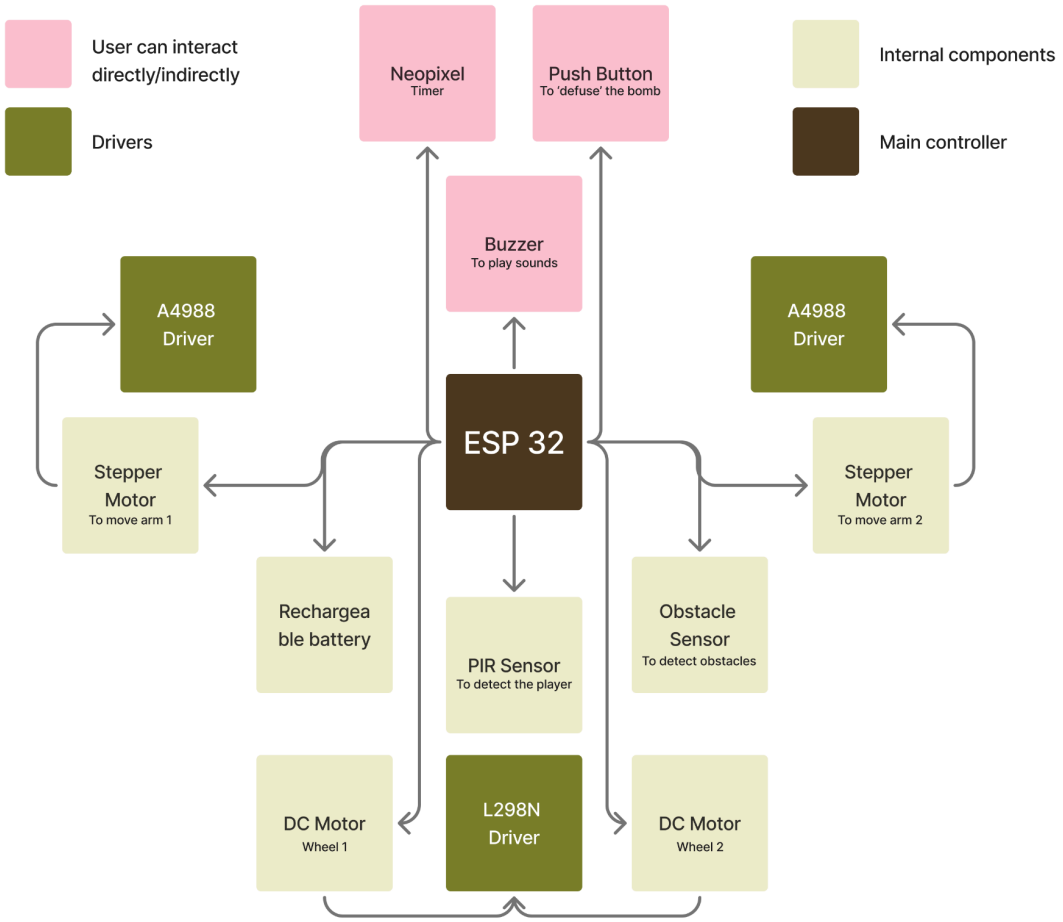
Mode	Trigger	Movement	Arms	Sound	NeoPixel
Wander	Player far away (PIR idle)	Slow random direction changes, occasional pauses	Lazy side-to-side wave	Slow quiet tick	Steady green → amber
Flee	Player within ~3m (PIR triggered)	Immediate random spin + full speed bolt	Frantic rapid flailing	Fast urgent tick	Pulsing amber → red
Victory	Button tagged in time	Stop all motors	Both arms raise and hold	Defuse fanfare	Green flash sequence
Explosion	Timer reaches zero	Stop all motors	Wild flailing then stop	Explosion sound	Rapid red flash

What are the main parts?

ESP 32- Main controller

2 DC Motors + Driver- For the wheels
2 Stepper Motors + 2 drivers- For each hand
PIR Sensor- For WALL-E to detect humans
Obstacle Sensor- To avoid bumping into walls, chairs, tables, etc
Buzzer- To play all required sounds
Push button- To defuse the "bomb"
Neopixel- To be used as the timer
LM2596 buck converter- To regulate the voltage.

6. High-Level Block Diagram



7. Input / Output Mapping

Input	Process	Output
User selects difficulty	App sends Bluetooth command (EASY / MEDIUM / HARD)	Robot sets timer duration & speed
User presses Start button	App sends "START" via Bluetooth	5-second countdown begins; NeoPixel flashes red then turns green
User presses Stop button	App sends "STOP" via Bluetooth	All motors halt immediately; system resets
Player is far away	Wall-E enters wander mode	Slow wandering movement
Player within ~3 m	PIR sensor triggered; Wall-E enters flee mode	Fast random spin & bolt; arms flailing
Player presses push button on robot	System checks if timer is still running	Victory dance and sound plays; NeoPixel turns green

8. Sketches & Concepts

· *Hand sketches / early ideas:*

- *A game where you have a small wrestling ring and the two characters in it are mario and princess peach. they are controlled and moved by esp32 components and mit app. They each have a light saber in their hands. The goal is for each player to successfully navigate their character from one end of the ring to the other person's base to win the game. If the players run into each other, whoever hits the other with the light saber first gets a point. so the one who gets either 5 points or reaches the base first wins.*
- *A paper robot that detects and picks up small waste using esp32 components whose direction is controlled via MIT app inventor*

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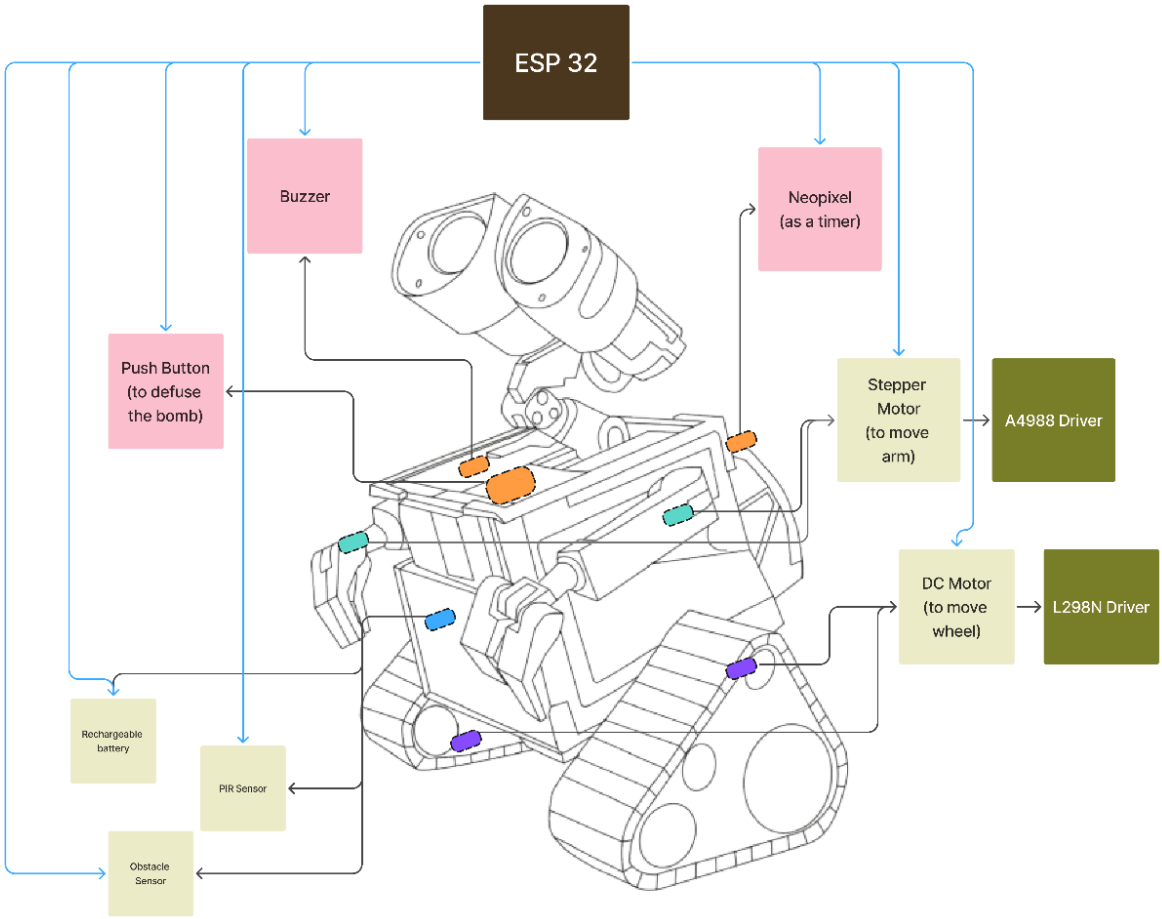
9. Final System Architecture (Labelled)

👉 *Clear diagram showing all components and connections*